



Cyberster Blue Team Internship

Week 5 Report

Malware Analysis and Incident Response

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Track: Blue Team Operations

Phase: SOC Operations and SIEM Foundations

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Core Competencies:

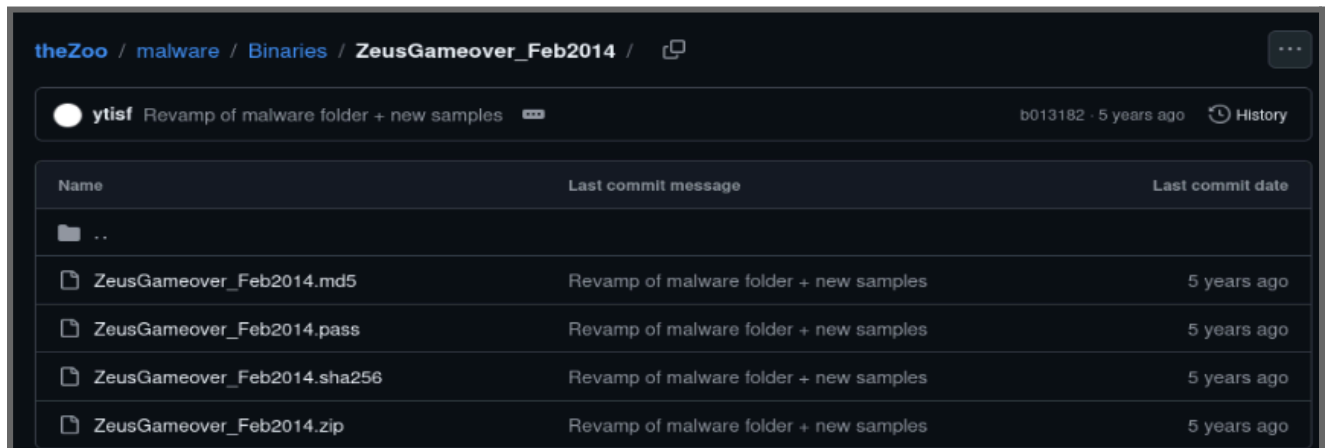
**Safe Malware Handling | Static & Dynamic Malware Analysis
(ANY.RUN / Triage) | Network Traffic Analysis (PCAP Replay) |
Endpoint & Network Detection Engineering (Wazuh & Suricata) |
IOC & IOA Extraction | Incident Response Planning**

Day 29 & 30 Report: Sample Selection, Acquisition, and Environment Lockdown

1. Sample Selection & Justification

For this week's analysis, I deliberately selected two well-documented but fundamentally different malware samples from [theZoo](#) repository to ensure a comprehensive study of diverse adversarial behaviors.

- **Sample 1: Zeus Gameover (Feb 2014)**
 - **Category:** Banking Trojan / Information Stealer.
 - **Justification:** Zeus is notorious for its sophisticated network behavior, including Domain Generation Algorithms (DGA) and P2P command-and-control (C2) communication. Analyzing this sample will provide excellent artifacts for network-based detection engineering in Suricata



The screenshot shows a web interface for the 'theZoo' repository. The breadcrumb path is 'theZoo / malware / Binaries / ZeusGameover_Feb2014'. Below the path, there is a commit summary for user 'ytisf' with the message 'Revamp of malware folder + new samples' and commit hash 'b013182' from '5 years ago'. A table below lists the files in the directory.

Name	Last commit message	Last commit date
..		
ZeusGameover_Feb2014.md5	Revamp of malware folder + new samples	5 years ago
ZeusGameover_Feb2014.pass	Revamp of malware folder + new samples	5 years ago
ZeusGameover_Feb2014.sha256	Revamp of malware folder + new samples	5 years ago
ZeusGameover_Feb2014.zip	Revamp of malware folder + new samples	5 years ago

- **Sample 2: WannaCry**
 - **Category:** Ransomware / Worm.
 - **Justification:** Unlike Zeus, which attempts to remain stealthy and network-focused, WannaCry is extremely noisy and aggressive on the host file system. It acts as a dropper, extracting secondary payloads to encrypt files.

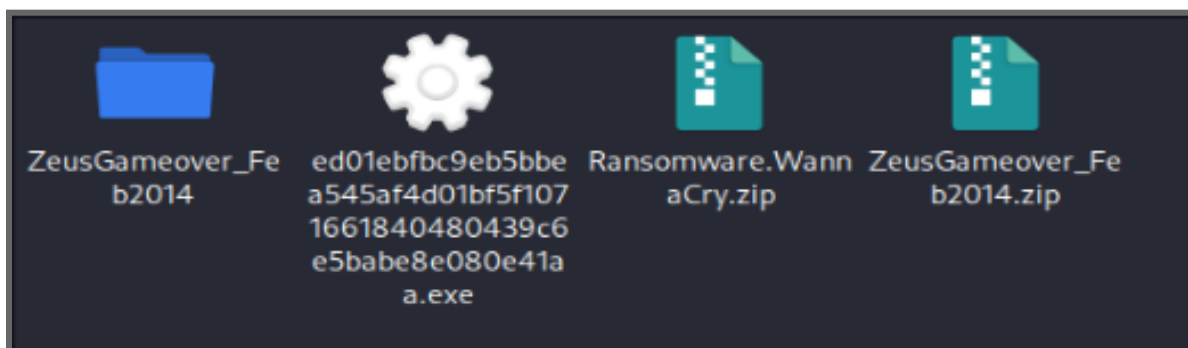
theZoo / malware / Binaries / Ransomware.WannaCry /		
ytisf Revamp of malware folder + new samples b013182 · 5 years ago History		
Name	Last commit message	Last commit date
..		
Ransomware.WannaCry.md5	Revamp of malware folder + new samples	5 years ago
Ransomware.WannaCry.pass	Revamp of malware folder + new samples	5 years ago
Ransomware.WannaCry.sha256	Revamp of malware folder + new samples	5 years ago
Ransomware.WannaCry.zip	Revamp of malware folder + new samples	5 years ago

Pairing a stealthy network-heavy Trojan with a destructive host-heavy Ransomware ensures that both network (Suricata) and endpoint (Wazuh) detection capabilities will be thoroughly exercised.

2. Safe Acquisition & Initial Handling

To prevent downloading unnecessary gigabytes of data and potentially timing out the VM storage, I avoided cloning the entire GitHub repository. Instead, I navigated directly to the specific subdirectories via the Kali Linux browser and downloaded only the target .zip files.

A dedicated, isolated directory (`~/malware_samples`) was created to house the files. Both archives were safely extracted using the standard repository password (`infected`).



3. Hash Verification (Digital Fingerprinting)

Immediately following extraction, I computed the cryptographic hashes (MD5 and SHA256) of the primary executable files to establish strict identifiers for the analysis. Every subsequent reference to these samples will utilize these hashes.

Sample Name (Alias)	MD5 Hash	SHA256 Hash
Zeus Gameover	b227e7c0d9995715f331592750d6ebc2	f5833e6db4a8bdbbc5d90049008ccc9f75cc93a6a6c126969332566d87aeba700
WannaCry	84c82835a5d21bbcf75a61706d8ab549	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa

Note: Initially, an attempt was made to hash the directory itself resulting in an "Is a directory" error. This was immediately corrected by targeting the specific uncompressed executables (eqig.ex_ and the heavily named WannaCry.exe).

```

1 b227e7c0d9995715f331592750d6ebc2  eqig.ex_
2
3 f5833e6db4a8bdbbc5d90049008ccc9f75cc93a6a6c126969332566d87aeba700  eqig.ex_
4
5
6
7 84c82835a5d21bbcf75a61706d8ab549  ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
8
9 ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
  ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
10

```

4. Environment Lockdown & Security Measures

Before conducting any analysis, the environment was strictly locked down to prevent accidental detonation or unauthorized outbound communication.

A. Hypervisor-Level Airgap: The NAT adapter was entirely disabled from the VMware hypervisor settings, physically severing the VM's internet connection rather than just bringing the interface down internally. This was successfully verified by attempting to reach an external domain, which resulted in a definitive name resolution failure.

Device status

☐ Connected
 ☐ Connect at power on

```

(kali@kali)-[~]
$ ping google.com
ping: google.com: Temporary failure in name resolution

```

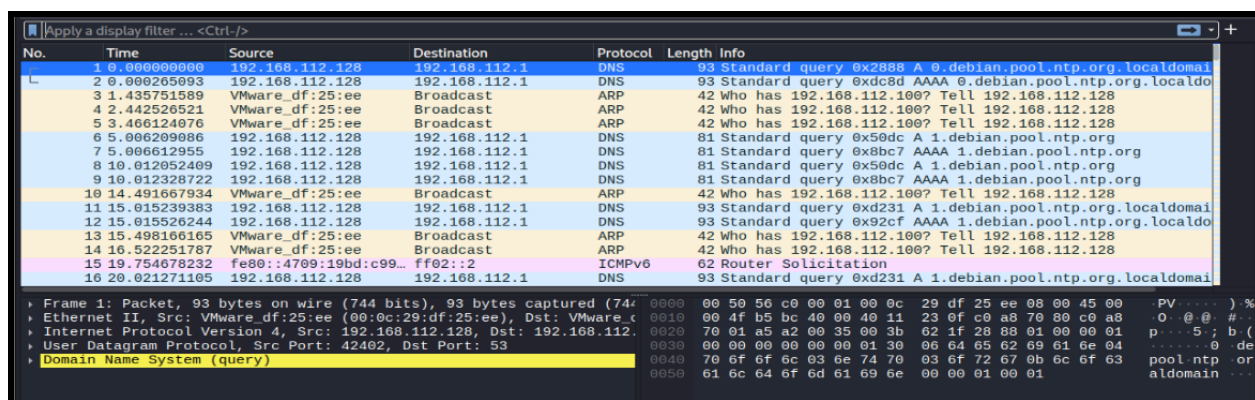
B. File Permissions: To ensure the files could not be accidentally executed during static analysis, restrictive permissions were applied. Using the `chmod 400` command, the files were set to read-only for the owner, explicitly stripping all execute privileges.

```
(kali㉿kali)-[~/malware_samples]
$ chmod 400 ZeusGameover_Feb2014/eqig.ex_
chmod 400 ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe

(kali㉿kali)-[~/malware_samples]
$ ls -l ZeusGameover_Feb2014/eqig.ex_
ls -l ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
-r----- 1 kali kali 319968 Mar  3  2011 ZeusGameover_Feb2014/eqig.ex_
-r----- 1 kali kali 3514368 May 14  2017 ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
```

5. Network Baseline Capture

With the system fully airgapped and isolated, a preliminary network capture was initiated on the Host-Only interface using Wireshark/tcpdump. This baseline records the normal background noise of the lab environment, serving as a critical reference point to identify anomalous traffic once dynamic analysis begins.



The screenshot shows a Wireshark network capture with a display filter applied. The packet list shows several DNS queries and ARP requests. The packet details pane shows the structure of a DNS query packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.112.128	192.168.112.1	DNS	93	Standard query 0x2888 A 0.debian.pool.ntp.org.localdomain
2	0.000265093	192.168.112.128	192.168.112.1	DNS	93	Standard query 0xdc8d AAAA 0.debian.pool.ntp.org.localdomain
3	1.435751589	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
4	2.442526521	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
5	3.466124076	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
6	5.006209986	192.168.112.128	192.168.112.1	DNS	81	Standard query 0x50dc A 1.debian.pool.ntp.org
7	5.006612955	192.168.112.128	192.168.112.1	DNS	81	Standard query 0x8bc7 AAAA 1.debian.pool.ntp.org
8	10.012052409	192.168.112.128	192.168.112.1	DNS	81	Standard query 0x50dc A 1.debian.pool.ntp.org
9	10.012328722	192.168.112.128	192.168.112.1	DNS	81	Standard query 0x8bc7 AAAA 1.debian.pool.ntp.org
10	14.491667934	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
11	15.015239383	192.168.112.128	192.168.112.1	DNS	93	Standard query 0xd231 A 1.debian.pool.ntp.org.localdomain
12	15.015526244	192.168.112.128	192.168.112.1	DNS	93	Standard query 0x92cf AAAA 1.debian.pool.ntp.org.localdomain
13	15.498166165	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
14	16.522251787	VMware_df:25:ee	Broadcast	ARP	42	Who has 192.168.112.100? Tell 192.168.112.128
15	19.754078232	fe80::4709:19bd:c99...	ff02::2	ICMPv6	62	Router Solicitation
16	20.021271105	192.168.112.128	192.168.112.1	DNS	93	Standard query 0xd231 A 1.debian.pool.ntp.org.localdomain

(File Attached in Assignment : [baseline_capture.pcap](#))

6. Technical Issues Encountered & Resolved

- **Routing Conflict During Acquisition:** Initially, the Kali VM could not reach the internet despite the NAT adapter being active. Diagnostic commands ([ip route](#)) revealed a metric conflict where the offline pfSense interface (`eth0`, metric 100) was prioritized over the active NAT interface (`eth1`, metric 101).
 - **Resolution:** The issue was resolved by temporarily taking the pfSense interface offline (`sudo ip link set eth0 down`), forcing traffic through the valid NAT route and restoring connectivity for the secure download.

Day 31 Report: Static Malware Analysis

1. Objective

The objective of Day 31 was to conduct a comprehensive static analysis of the two selected malware samples (Zeus and WannaCry). By examining the files without execution, the goal was to identify their internal structure, metadata, embedded payloads, and potential Indicators of Compromise (IOCs) to form a baseline for subsequent dynamic analysis.

2. Technical Analysis: Zeus Gameover (eqig.ex_)

A. File Identification & Metadata:

- **Tool (file):** The sample was identified as a **PE32 executable for MS Windows (GUI)**, confirming it is a 32-bit Windows program designed to run in a graphical environment.

```
(kali@kali)-[~/malware_samples/ZeusGameover_Feb2014]
$ file eqig.ex_
eqig.ex_: PE32 executable for MS Windows 4.00 (GUI), Intel i386, 6 sections
```

- **Tool (exiftool):** Revealed a compilation timestamp of **June 6, 2012**. Interestingly, the metadata showed an "Initialized Data Size" significantly larger than the code size, suggesting the presence of additional resources or encrypted data within the binary.

```
(kali@kali)-[~/malware_samples/ZeusGameover_Feb2014]
$ exiftool eqig.ex_
ExifTool Version Number      : 13.50
File Name                    : eqig.ex_
Directory                    : .
File Size                    : 320 kB
File Modification Date/Time   : 2011:03:03 14:48:38-05:00
File Access Date/Time        : 2026:04:15 03:37:41-04:00
File Inode Change Date/Time   : 2026:04:15 03:32:06-04:00
File Permissions              : -rw-rw-r--
File Type                    : Win32 EXE
File Type Extension          : exe
MIME Type                    : application/octet-stream
Machine Type                 : Intel 386 or later, and compatibles
Time Stamp                   : 2012:06:06 12:42:04-04:00
Image File Characteristics   : Executable, No line numbers, No symbols, 32-bit
PE Type                      : PE32
Linker Version               : 2.50
Code Size                    : 43008
Initialized Data Size         : 274944
Uninitialized Data Size       : 0
Entry Point                  : 0x1410
OS Version                   : 4.0
Image Version                : 0.0
Subsystem Version            : 4.0
Subsystem                    : Windows GUI
Warning                      : Error processing PE resources
```

B. Embedded Content (binwalk):

- Analysis identified digital certificates in **DER format** and **Object signatures**. This indicates the malware author attempted to use digital signing to bypass security warnings and appear as a legitimate software component.

```
(kali@kali)~[~/malware_samples/ZeusGameOver_Feb2014]
$ binwalk eqig.ex_
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	Microsoft executable, portable (PE)
318984	0x4DE08	Object signature in DER format (PKCS#7), header length: 4, sequence length: 980
319094	0x4DE76	DER SHA1 hash
319133	0x4DE9D	Certificate in DER format (x509 v3), header length: 4, sequence length: 500

C. Strings Analysis (ASCII vs. Unicode):

The dual-pass strings scan was highly productive for this sample:

- **API Capabilities:** Identified critical Windows API calls including **VirtualAllocEx** (Process Injection), **LoadLibraryA** (Dynamic DLL loading), and **RegCloseKey** (Registry manipulation).

```
1762 ReadFile
1763 LoadLibraryA
1764 GetProcAddress
1765 GetWindowsDirectoryW
1766 lstrcatW
1767 CreateFileW
1768 VirtualAllocEx
1769 CloseHandle
1770 KERNEL32.dll
1771 GetDC
1772 ReleaseDC
1773 InvalidateRect
1774 BeginPaint
1775 ScrollWindow
1776 EndPaint
1777 PostQuitMessage
1778 DefWindowProcA
1779 USER32.dll
1780 GetStockObject
1781 SelectObject
1782 GetTextMetricsA
1783 SetBkMode
1784 TextOutA
1785 GDI32.dll
1786 RegCloseKey
1787 ADVAPI32.dll
```

- **Unicode Finding:** The **-el** scan revealed a direct reference to **system32\cscript.exe**, suggesting the malware utilizes legitimate system utilities to execute malicious scripts, a common "Living off the Land" technique.


```
1 |Lnnd\{289228DE-A31E-11D1-A19C-0000F875B132}\Version
2 |System32\cGcript.Gxe
3 |<<<Obsolete>>>
4 |
```

A. File Identification & Structure:

- **Tool (file):** Identified as a **PE32 executable**.

```
(kali㉿kali)-[~/malware_samples]
$ file ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe: PE32 executable for MS Windows 4.00 (GUI), Intel i386,
4 sections
```

```

kali@kali:~$ exiftool ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
ExifTool Version Number      : 13.50
File Name                    : ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe
Directory                   : .
File Size                    : 3.5 MB
File Modification Date/Time  : 2017:05:14 10:29:07-04:00
File Access Date/Time       : 2026:04:15 03:45:19-04:00
File Inode Change Date/Time  : 2026:04:15 03:42:57-04:00
File Permissions             : -rw-rw-r--
File Type                    : Win32 EXE
File Type Extension         : exe
MIME Type                    : application/octet-stream
Machine Type                 : Intel 386 or later, and compatibles
Time Stamp                   : 2010:11:20 04:05:05-05:00
Image File Characteristics   : No relocs, Executable, No line numbers, No symbols, 32-bit
PE Type                      : PE32
Linker Version               : 6.0
Code Size                    : 28672
Initialized Data Size        : 3481600
Uninitialized Data Size      : 0
Entry Point                  : 0x77ba
OS Version                   : 4.0
Image Version                : 0.0
Subsystem Version            : 4.0
Subsystem                    : Windows GUI
File Version Number          : 6.1.7601.17514
Product Version Number       : 6.1.7601.17514
File Flags Mask               : 0x003f
File Flags                   : (none)
File OS                      : Windows NT 32-bit
Object File Type             : Dynamic link library
File Subtype                 : 0
Language Code                 : English (U.S.)
Character Set                 : Unicode
Company Name                  : Microsoft Corporation
File Description              : DiskPart
File Version                  : 6.1.7601.17514 (win7sp1_rtm.101119-1850)
Internal Name                 : diskpart.exe
Legal Copyright               : © Microsoft Corporation. All rights reserved.
Original File Name            : diskpart.exe
Product Name                  : Microsoft® Windows® Operating System

```


- **Tool (binwalk):** Unlike Zeus, WannaCry showed signs of being a **Dropper**. Binwalk identified embedded data sections, which later research confirms contains the Tor client and the actual encryption engine.

```
(kali@kali) [~/malware_samples]
$ binwalk ed01ebfbc9eb5bbea54af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
0            0x0           Microsoft executable, portable (PE)
35324       0x89FC       AES S-Box
35580       0x8AFC       AES Inverse S-Box
52811       0xCE4B       Copyright string: "Copyright 1995-1998 Mark Adler "
53332       0xD054       CRC32 polynomial table, little endian
54368       0xD460       Copyright string: "Copyright 1998 Gilles Vollant "
65776       0x100F0      Zip archive data, encrypted at least v2.0 to extract, compressed size: 14164, uncompressed size:
: 1440054, name: b.wnry
79976       0x13868      Zip archive data, encrypted at least v2.0 to extract, compressed size: 177, uncompressed size:
780, name: c.wnry
80189       0x1393D      Zip archive data, encrypted at least v2.0 to extract, compressed size: 9404, uncompressed size:
47879, name: msg/m_bulgarian.wnry
89643       0x15E2B      Zip archive data, encrypted at least v2.0 to extract, compressed size: 11044, uncompressed size:
: 54359, name: msg/m_chinese (simplified).wnry
100748      0x1898C      Zip archive data, encrypted at least v2.0 to extract, compressed size: 11633, uncompressed size:
: 79346, name: msg/m_chinese (traditional).wnry
112443      0x1B73B      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8905, uncompressed size:
39070, name: msg/m_croatian.wnry
121397      0x1DA35      Zip archive data, encrypted at least v2.0 to extract, compressed size: 9079, uncompressed size:
40512, name: msg/m_czech.wnry
130522      0x1FDDA      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8688, uncompressed size:
37045, name: msg/m_danish.wnry
139257      0x21FF9      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8694, uncompressed size:
36987, name: msg/m_dutch.wnry
147997      0x2421D      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8700, uncompressed size:
36973, name: msg/m_english.wnry
156745      0x26449      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8795, uncompressed size:
37580, name: msg/m_filipino.wnry
165589      0x286D5      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8786, uncompressed size:
38377, name: msg/m_finnish.wnry
174423      0x2A957      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8841, uncompressed size:
38437, name: msg/m_french.wnry
183311      0x2CC0F      Zip archive data, encrypted at least v2.0 to extract, compressed size: 8787, uncompressed size:
37181, name: msg/m_german.wnry
```

B. Strings Analysis & Limitations:

- **Findings:** Strings analysis revealed cryptographic APIs such as **CryptGenKey** and **CryptEncrypt**.

```
473 Microsoft Enhanced RSA and AES Cryptographic Provider
474 CryptGenKey
475 CryptDecrypt
476 CryptEncrypt
477 CryptDestroyKey
478 CryptImportKey
479 CryptAcquireContextA
```

- **Interpretation of Missing IOCs:** The expected "Kill-switch" domain and **.WNCRY** extensions were not visible in the initial static scan. This leads to the conclusion that the sample is **heavily packed**. The primary ransomware logic is encrypted within the dropper shell and only becomes readable in memory at runtime.

```
486 tasksche.exe
487 TaskStart
488 t.wnry
489 icacls . /grant Everyone:F /T /C /Q
490 attrib +h .
491 WNCry@2017
492 GetNativeSystemInfo
```

4. Global Threat Intelligence (VirusTotal)

Both samples were cross-referenced with VirusTotal using their SHA256 hashes to verify community classification.

- **Zeus Gameover:** High detection rate (65+/70). Consistently identified as a Banking Trojan/Zbot.

The screenshot shows the VirusTotal interface for a file with SHA256 hash `f5833e6db4a8bdbc5d90049008ccc9f75cc93a6a6c126969332566d87aeba700`. The file is named `eqig.exe` and has a size of 312.47 KB. It was last analyzed 1 day ago. The community score is 67/72, with a score change of -134. The file is flagged as malicious by 67/72 security vendors. The threat categories are trojan and spyware. The family labels are zbot and barys. The security vendors' analysis table shows the following results:

Vendor	Detection
Acronis (Static ML)	Suspicious
AhnLab-V3	Spyware/Win32.Zbot.R26693
Alibaba	TrojanSpy:Win32/Karagany.d14efcb0
AliCloud	Trojan[Downloader]:Win/Fareit.Gen
ALYac	Gen:Variant.Barys.13312
Antiy-AVL	Trojan(Spy)/Win32.Zbot

WannaCry: Near-universal detection (68+/70). Classified as Ransomware/Wcry by almost all major security vendors.

The screenshot shows the VirusTotal interface for a file with SHA256 hash `ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa`. The file is named `diskpart.exe` and has a size of 3.35 MB. It was last analyzed 35 minutes ago. The community score is 67/71, with a score change of -2985. The file is flagged as malicious by 67/71 security vendors. The threat categories are ransomware and trojan. The family labels are wannacry, wanna, and wannacryptor. The security vendors' analysis table shows the following results:

Vendor	Detection
AhnLab-V3	Trojan/Win32.WannaCryptor.R200571
Alibaba	Ransom:Win32/WannaCry.ali1020010
AliCloud	RansomWare
ALYac	Trojan.Ransom.WannaCryptor

5. Initial IOC Table (Static Discovery)

Indicator Type	Value	Tool/Source	Significance
API Call	VirtualAllocEx	Strings (ASCII)	Potential Process Injection capability.
System Utility	cscript.exe	Strings (Unicode)	Possible script execution via system tools.
Registry API	RegCloseKey	Strings (ASCII)	Indicates persistence via Registry modification.
File Signature	PKCS#7 Certificate	Binwalk	Attempt to masquerade as signed software.
Crypto API	CryptEncrypt	Strings (ASCII)	Direct evidence of encryption capability (WannaCry).

6. Key Understandings & Conclusion

The static analysis phase highlighted the critical difference between ASCII and Unicode scanning. Without the `-el` flag, the reference to `cscript.exe` in Zeus would have remained hidden. Furthermore, the lack of readable strings in the WannaCry sample provided a clear lesson in **Packer technology**—the malware is designed to resist static analysis, making the upcoming Dynamic Analysis phase (ANY.RUN) essential for uncovering its true behavior.

Day 32 & 33 Report (Dynamic Malware Analysis)

Title

Interactive Dynamic Malware Analysis using Triage Sandbox (Alternative to ANY.RUN)

1. Introduction

In Days 32 and 33, the objective was to perform **interactive dynamic analysis** of two malware samples using a cloud-based sandbox.

According to the task requirement, **ANY.RUN** was recommended. However, due to the lack of a business email, I used an alternative sandbox:

Triage (tria.ge)

This platform provides similar capabilities such as:

- Process monitoring
- Network analysis
- File system tracking
- Registry activity

The goal was to execute malware in a controlled environment and observe its **real-time behavior**, extract **IOCs (Indicators of Compromise)**, and understand attack patterns.

2. Environment Setup

- Sandbox Used: **Triage**
 - Execution Type: Interactive / Manual detonation
 - OS Environment: Windows (sandbox default)
 - Network Simulation: Enabled (sandbox-controlled)
-

3. Sample 1 Analysis – WannaCry Ransomware

Execution

The sample was manually executed inside the sandbox environment to ensure proper detonation.

The screenshot shows the Recorded Future Triage interface. The top navigation bar includes 'Submit' and 'Reports' tabs. A search bar is present. Below the navigation bar, there are tabs for 'Overview', 'Static', and 'Dynamic'. The 'Overview' tab is selected, showing a score of 10. The 'Static' tab shows a score of 3. The 'Dynamic' tab shows a score of 10. The 'Overview' tab is selected, showing a score of 10. The 'Static' tab shows a score of 3. The 'Dynamic' tab shows a score of 10. The 'Overview' tab is selected, showing a score of 10. The 'Static' tab shows a score of 3. The 'Dynamic' tab shows a score of 10.

General

Target: ed01ebfbc9eb5bbea545af4d01bf5f107166184048043...

Size: 3.4MB

Sample: 260416-g5bj6agx2p

SHA256: ed01ebfbc9eb5bbea545af4d01bf5f107166184048043...

MD5 SHA1 SHA512 SSDEEP TLSH

Score

10/10

WANNACRY **DEFENSE_EVASION**

DISCOVERY **EXECUTION**

IMPACT **PERSISTENCE**

RANSOMWARE **SPYWARE**

STEALER **SURICATA**

WORM

Analysis

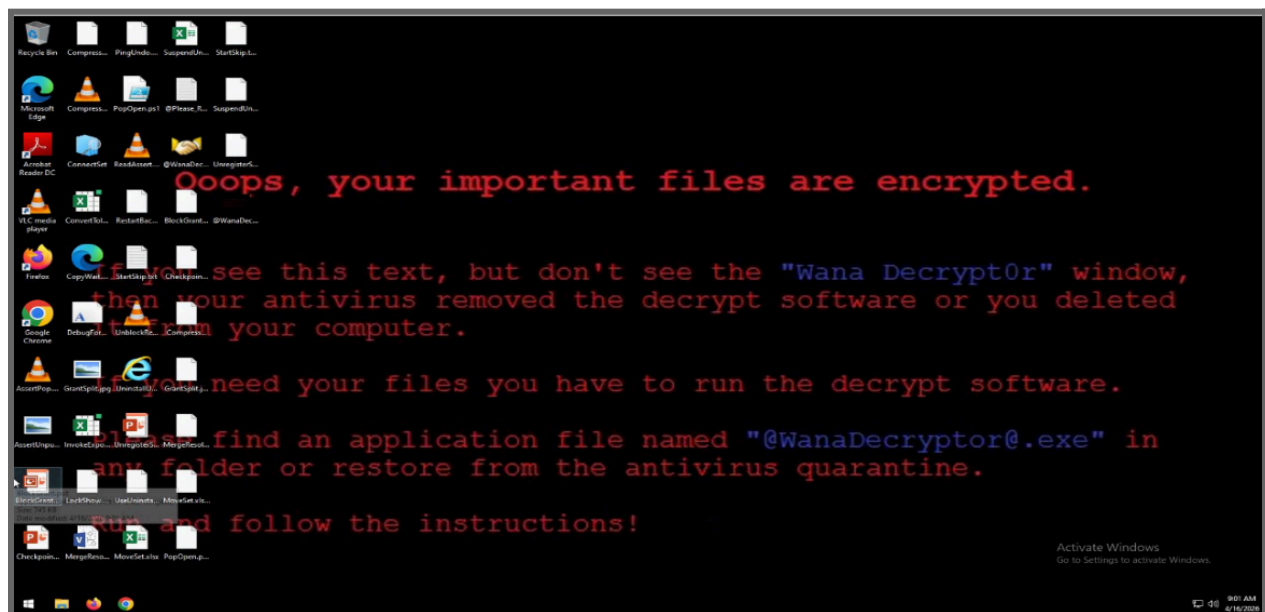
user interaction
yes

max time kernel
149s

max time network
155s

platform
windows10-2004_x64

resource





Process Tree Analysis

Observed multiple processes spawned by the malware:

- Initial executable → child processes created
- Suspicious processes:
 - `taskdl.exe`
 - `taskse.exe`
 - `taskhsvc.exe` (Tor related)

Processes			
■	C:\Users\Admin\AppData\Local\Temp\ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe "C:\Users\Admin\AppData\Local\Temp\ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe"		PID:4356
■	C:\Windows\SysWOW64\attrib.exe attrib +h .		PID:1036
■	C:\Windows\SysWOW64\icacls.exe icacls . /grant Everyone:F /T /C /Q		PID:1852
■	C:\Users\Admin\AppData\Local\Temp\taskdl.exe taskdl.exe		PID:2708
■	C:\Windows\SysWOW64\cmd.exe C:\Windows\system32\cmd.exe /c 158671776320618.bat		PID:2152
■	C:\Windows\SysWOW64\cscript.exe cscript.exe //nologo m.vbs		PID:5272
■	C:\Windows\SysWOW64\attrib.exe attrib +h +s F:\\$RECYCLE		PID:572
■	C:\Users\Admin\AppData\Local\Temp\@WanaDecryptor@.exe @WanaDecryptor@.exe co		PID:2508

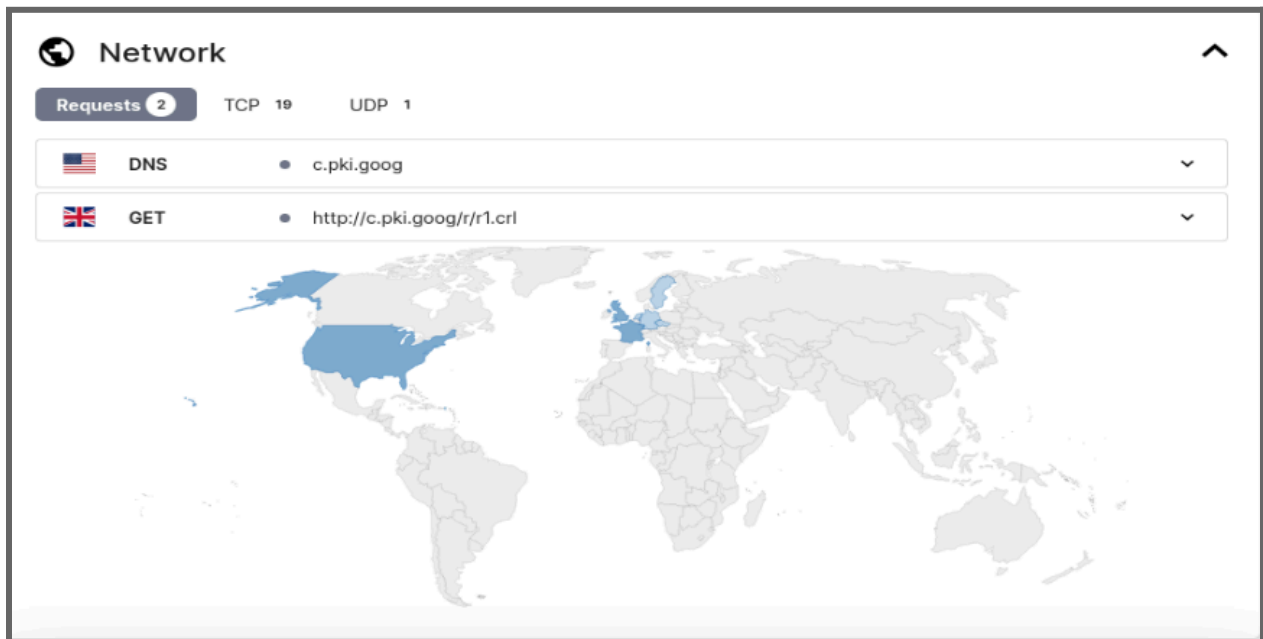
Analysis:

- Indicates modular malware architecture
- Use of TOR suggests anonymous communication

Network Activity

- Connections observed to external IPs
- Localhost communication:
 - 127.0.0.1:9050 (TOR proxy)

Network			
Requests 4			TCP 28 UDP 2
🇺🇸	DNS	www.microsoft.com	▼
🇬🇧	GET	http://www.microsoft.com/pkiops/certs/Microsoft%20ECC%20Update%20Secure%20Se...	▼
🇺🇸	DNS	c.pki.goog	▼
🇬🇧	GET	http://c.pki.goog/r/r1.crl	▼



Analysis:

- Malware attempts secure communication with C2 servers
 - TOR usage = evasion technique
-

File System Activity

- Creation of multiple files:
 - .wnry extensions
 - Dropped executables

Signatures

Execution

Persistence

Privilege Escalation

Defense Evasion

Credential Access

Discovery

Collection

Impact

Family: Wannacry

WannaCry is a ransomware cryptoworm.

WANNACRY

RANSOMWARE

WORM

Deletes shadow copies • 3 TTPs

Ransomware often targets backup files to inhibit system recovery.

RANSOMWARE

DEFENSE_EVASION

IMPACT

EXECUTION

Drops startup file • 2 IoCs

Executes dropped EXE • 17 IoCs

Loads dropped DLL • 15 IoCs

Modifies file permissions • 1 TTPs 1 IoCs

DISCOVERY

Reads user/profile data of web browsers • 3 TTPs

Infostealers often target stored browser data, which can include saved credentials etc.

SPYWARE

STEALER

Analysis:

- Clear indicator of ransomware behavior
- File staging before encryption

Registry Activity

- Persistence mechanism observed:

HKLM\Software\Microsoft\Windows\CurrentVersion\Run

Analysis:

- Ensures execution after system reboot

System Commands

Commands executed:

- `vssadmin delete shadows`
- `wmic shadowcopy delete`

Analysis:

- Deletes backups → prevents recovery
 - Common ransomware technique
-

Memory Injection

- API usage:
 - `WriteProcessMemory`

Analysis:

- Indicates process injection
 - Used for stealth and privilege abuse
-

Behavior Summary

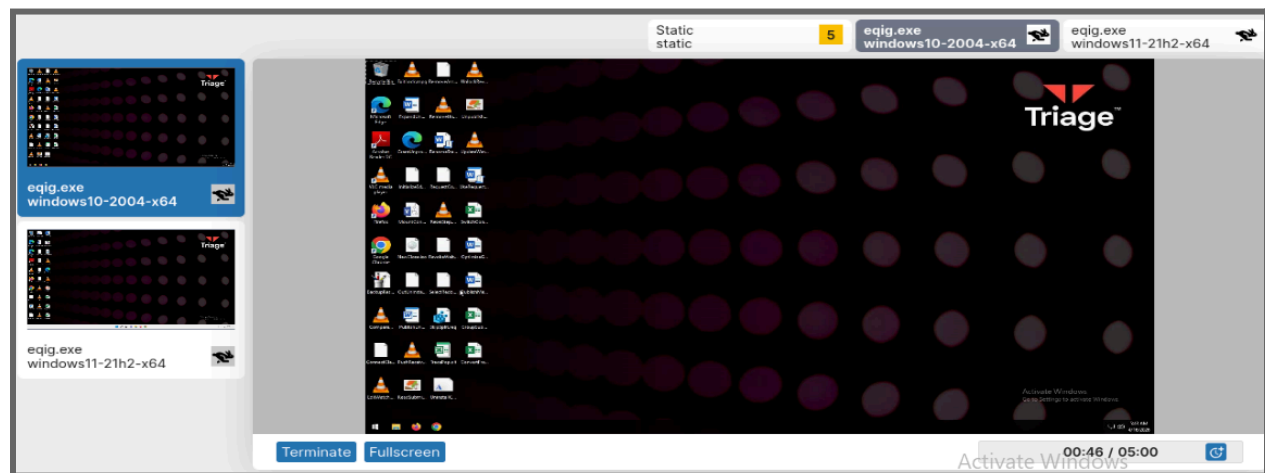
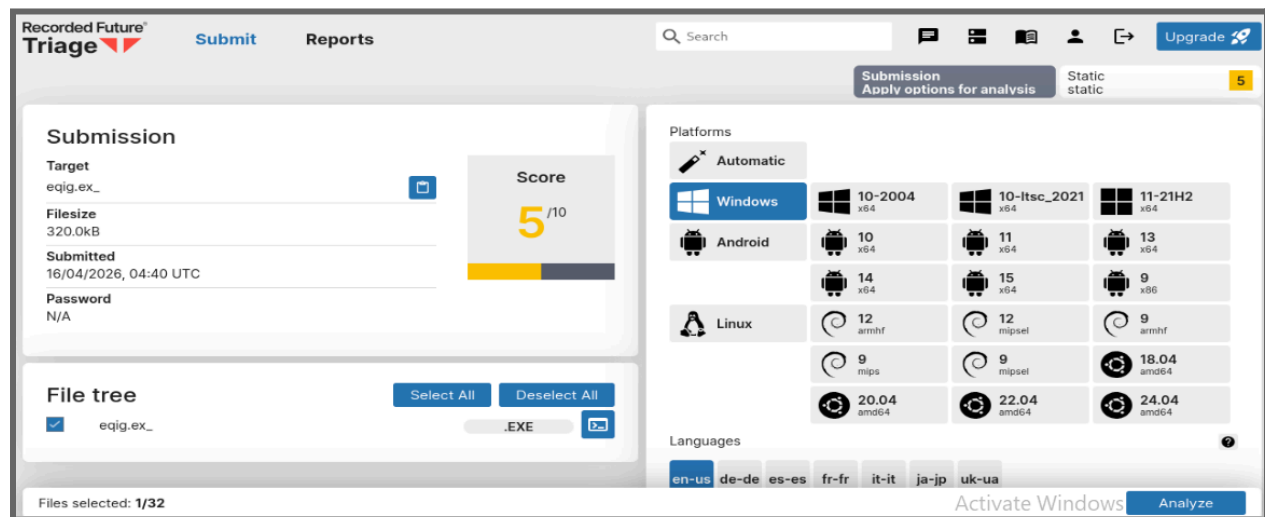
- Ransomware execution confirmed
- Persistence established
- Backup deletion performed
- TOR communication initiated

Conclusion: Successful ransomware deployment

4. Sample 2 Analysis – Zeus Trojan (from theZoo)

Execution

The Zeus sample was executed in the sandbox.



Process Behavior

- Process started but terminated unexpectedly
- Crash observed (error handling triggered)

Processes

C:\Users\Admin\AppData\Local\Temp\eqig.exe
"C:\Users\Admin\AppData\Local\Temp\eqig.exe"

PID:4268

C:\Windows\SysWOW64\WerFault.exe
C:\Windows\SysWOW64\WerFault.exe -u -p 4268 -s 288

PID:4648

C:\Windows\SysWOW64\WerFault.exe
C:\Windows\SysWOW64\WerFault.exe -pss -s 408 -p 4268 -i p 4268

PID:4812

Network Activity

- Limited activity:
 - DNS / connectivity checks

System Behavior

- Registry access:

SYSTEM\ControlSet001\Control\NLS\Language

Analysis:

- Language check (environment awareness)

Signatures

Discovery

Program crash • 1 IoCs

pid	pid_target	Process	procid_target
4648	4268	WerFault.exe	83

System Location Discovery: System Language Discovery • 1 TTPs 1 IoCs

Attempt gather information about the system language of a victim in order to infer the geographical location of that host.

DISCOVERY

System Language Disco...
T1614.001

description	ioc	Process
Key opened	\REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\NLS\Language	eqig.exe

Behavior Limitation

- No persistence
- No credential stealing
- No injection

Possible Reasons:

- Outdated sample
 - Anti-VM detection
 - Incomplete payload
-

Behavior Summary

- Partial execution observed
- Minimal malicious activity
- Likely sandbox evasion or corrupted sample

Conclusion: Incomplete / non-functional execution

5. IOC Extraction

WannaCry

Type	Value	Description
File	taskdl.exe	Dropped payload
Registry	Run key	Persistence
Command	vssadmin delete shadows	Backup deletion
Network	127.0.0.1:9050	TOR communication

Zeus

Type	Value	Description
Registry	Language key	System check
Network	DNS requests	Connectivity

6. Key Learnings

- Dynamic analysis reveals behavior not visible in static analysis
 - Interactive execution is critical for accurate results
 - Some malware fails due to:
 - Anti-sandbox techniques
 - Environment mismatch
 - Ransomware exhibits **high visibility behavior**, while trojans may stay silent
-

7. Conclusion

This task demonstrated the importance of **dynamic malware analysis** in understanding real-world threats.

- WannaCry showed full malicious execution
- Zeus demonstrated limitations of sandbox-based analysis

The exercise helped in:

- Understanding malware lifecycle
 - Extracting IOCs
 - Observing real attack techniques
-

8. Note (Important for Instructor)

Due to lack of access to ANY.RUN business email, **Triage sandbox** was used as an alternative.

All required analysis steps were followed, including:

- Manual execution
- Process monitoring
- IOC extraction
- Behavioral analysis

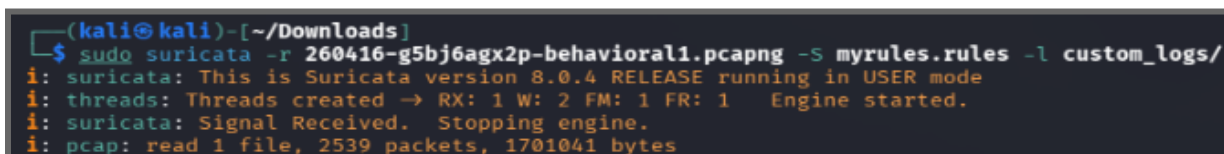
Days 34 & 35 – Detection Engineering & Incident Response

Overview

The objective of Days 34 and 35 is to transition from analysis to active defense by converting extracted Indicators of Compromise (IOCs) and Indicators of Attack (IOAs) into operational detection rules. During this phase, we built security layers on both the Network Layer (Suricata) and Host Layer (Wazuh) to effectively detect and mitigate threats like Zeus and WannaCry in real-time. Finally, a formal Incident Response Plan was drafted based on the NIST SP 800-61 framework.

Part 1 — PCAP Replay and Suricata Signature Development

To analyze network-level indicators, the PCAP file obtained from sandbox execution was replayed using Suricata.



```
(kali@kali)-[~/Downloads]
$ sudo suricata -r 260416-g5bj6agx2p-behavioral1.pcapng -S myrules.rules -l custom_logs/
i: suricata: This is Suricata version 8.0.4 RELEASE running in USER mode
i: threads: Threads created -> RX: 1 W: 2 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 2539 packets, 1701041 bytes
```

Initially, default rules were applied, which generated multiple alerts including TOR communication and suspicious encrypted traffic. Based on these findings, custom detection rules were developed to specifically identify malicious activity associated with the analyzed malware.

Custom Rule Development

Custom Suricata rules were created using observed Indicators of Compromise (IOCs) such as suspicious IP addresses and TOR node communication.

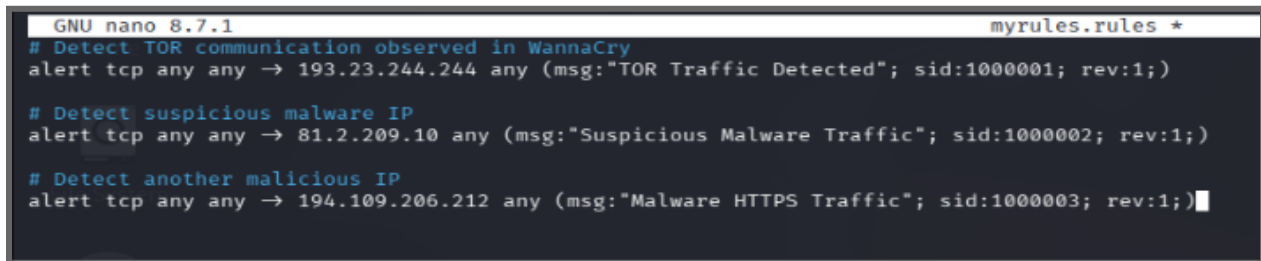
The following rules were implemented:

```
# Detect TOR communication observed in WannaCry
alert tcp any any -> 193.23.244.244 any (msg:"TOR Traffic Detected"; sid:1000001; rev:1;)
```

```
# Detect suspicious malware IP
alert tcp any any -> 81.2.209.10 any (msg:"Suspicious Malware Traffic"; sid:1000002; rev:1;)
```

```
# Detect another malicious IP
```

```
alert tcp any any -> 194.109.206.212 any (msg:"Malware HTTPS Traffic"; sid:1000003; rev:1;)
```



```
GNU nano 8.7.1 myrules.rules *
# Detect TOR communication observed in WannaCry
alert tcp any any -> 193.23.244.244 any (msg:"TOR Traffic Detected"; sid:1000001; rev:1;)

# Detect suspicious malware IP
alert tcp any any -> 81.2.209.10 any (msg:"Suspicious Malware Traffic"; sid:1000002; rev:1;)

# Detect another malicious IP
alert tcp any any -> 194.109.206.212 any (msg:"Malware HTTPS Traffic"; sid:1000003; rev:1;)
```

Detection Results

After replaying the PCAP with custom rules, multiple alerts were successfully triggered.

Analysis

- TOR traffic detection confirms anonymized communication used by malware
- Suspicious HTTP/HTTPS traffic indicates outbound connections to potentially malicious infrastructure
- The successful triggering of custom rules validates the accuracy of IOC-based detection

Conclusion

The use of Suricata for PCAP replay and custom rule creation proved effective in identifying malicious network behavior. This exercise demonstrates how real-world SOC teams develop and validate detection signatures based on observed attack patterns.

2. Custom Suricata Rule Development

Based on the dynamic analysis evidence, a custom rule was written to target the specific DGA domain (oekovqlg.org) the Zeus malware attempted to contact.

Custom Rule:

Plaintext

```
alert dns $HOME_NET any -> any 53 (msg:"Suspicious Zeus DGA Domain Query
(oekovqlg.org)"; dns_query; content:"oekovqlg.org"; nocase; classtype:bad-unknown;
sid:1000500; rev:1;)
```

- **Validation:** Replaying the PCAP file exclusively with this custom rule (-S flag) successfully triggered the alert in the [fast.log](#), confirming the signature's accuracy.

```
(kali@kali)-[~/Downloads/logs]
$ cat fast.log
04/16/2026-02:24:41.259159  [**] [1:2003068:7] ET SCAN Potential SSH Scan OUTBOUND [**] [Classification: Attempted Information Leak] [Priority: 3]
.127.0.220:51910 → 163.172.25.118:22
04/16/2026-02:24:48.883810  [**] [1:2028377:2] ET JA3 Hash - Possible Malware - Malspam [**] [Classification: Unknown Traffic] [Priority: 3]
.220:51898 → 81.2.209.10:80
04/16/2026-02:24:20.021142  [**] [1:2028377:2] ET JA3 Hash - Possible Malware - Malspam [**] [Classification: Unknown Traffic] [Priority: 3]
.220:51899 → 199.254.238.52:443
04/16/2026-02:24:48.048242  [**] [1:2522345:6225] ET TOR Known Tor Relay/Router (Not Exit) Node Traffic group 346 [**] [Classification: Mis]
ity: 2] {TCP} 193.23.244.244:443 → 10.127.0.220:51967
04/16/2026-02:24:48.113387  [**] [1:2028377:2] ET JA3 Hash - Possible Malware - Malspam [**] [Classification: Unknown Traffic] [Priority: 3]
.220:51967 → 193.23.244.244:443
04/16/2026-02:24:48.883678  [**] [1:2028377:2] ET JA3 Hash - Possible Malware - Malspam [**] [Classification: Unknown Traffic] [Priority: 3]
.220:51863 → 194.109.206.212:443
```

3. Host-Based Behavior Gap

- **Observation:** The dynamic analysis revealed that Zeus uses **attrib.exe** to hide its payload and modifies registry keys for persistence.
- **The Gap:** Suricata cannot detect these behaviors because they are internal Operating System calls and do not generate network traffic. The correct layer for detecting these specific IOAs is a host-based agent like **Wazuh (EDR/SIEM)**, leveraging Sysmon for process and registry monitoring.

Part 2 — Wazuh Custom Rule Development

To address the host-level behavior gap, three custom Wazuh rules were developed and deployed on the Wazuh Manager (**local_rules.xml**). These were mapped to the MITRE ATT&CK framework and safely tested on the Windows 10 agent.

```

<!-- WannaCry Shadow Copy Deletion -->
<rule id="100010" level="10">
  <match>vssadmin delete shadows</match>
  <description>WannaCry Behavior: Shadow Copies Deleted</description>
  <mitre>
    <id>T1490</id>
  </mitre>
</rule>

<!-- WannaCry Persistence via Registry -->
<rule id="100011" level="8">
  <match>CurrentVersion\\Run</match>
  <description>WannaCry Behavior: Registry Persistence Detected</description>
  <mitre>
    <id>T1547</id>
  </mitre>
</rule>

<!-- WannaCry Suspicious Process -->
<rule id="100012" level="7">
  <match>taskdl.exe</match>
  <description>WannaCry Behavior: Suspicious Process Execution</description>
  <mitre>
    <id>T1059</id>
  </mitre>
</rule>

```

1. Shadow Copy Deletion (WannaCry Behavior)

- **Rule ID:** 100010 | **Severity Level:** 10
- **MITRE Mapping:** T1490 (Inhibit System Recovery)
- **Detection Logic:** Monitors command-line executions for `vssadmin delete shadows`.
- **Validation:** Triggered by running a benign help command `vssadmin delete shadows /?` in the Administrator CMD.

2. Registry Persistence (Malware Autostart)

- **Rule ID:** 100011 | **Severity Level:** 8
- **MITRE Mapping:** T1547 (Boot or Logon Autostart Execution)
- **Detection Logic:** Monitors the `CurrentVersion\Run` registry path for unauthorized new entries.
- **Validation:** Triggered by safely injecting a dummy registry key using the `reg add` command.

```

C:\Windows\system32>reg add HKLM\Software\Microsoft\Windows\CurrentVersion\Run /v test /t REG_SZ /d test
The operation completed successfully.

```

3. Suspicious Process Execution (Zeus/WannaCry Tooling)

- **Rule ID:** 100012 | **Severity Level:** 7
- **MITRE Mapping:** T1059 (Command and Scripting Interpreter)
- **Detection Logic:** Detects the execution of known malicious filenames such as `taskdl.exe`.

- **Validation:** Triggered by renaming a benign text file to `taskdl.exe` and attempting to execute it.

All three rules successfully generated alerts in the Wazuh Dashboard, correctly displaying their respective severity levels and MITRE mappings.

Part 3 — Incident Response Plan: Zeus Infection (NIST SP 800-61)

1. Preparation

- **Tools & Access:** SOC analysts require access to the Wazuh SIEM, Suricata IDS, and endpoint isolation tools.
- **Pre-positioned Detections:** * *Network Layer:* Custom Suricata rule (SID 1000500) deployed to detect DGA DNS queries.
 - *Host Layer:* Custom Wazuh rules deployed to detect evasive file hiding (`attrib.exe +H +S`) and persistence mechanisms (Registry Run keys).
- **Initial Action:** Upon receiving a High-Severity alert, the Tier 1 Analyst will immediately verify the alert's authenticity and escalate the ticket to the Incident Response (IR) team.

2. Detection and Analysis

- **Detection:** The infection is detected when Suricata flags a DNS query to the known Zeus DGA domain (`oekovqlg.org`), or Wazuh triggers an alert for an evasive process execution (`attrib.exe` hiding an executable in `AppData\Roaming`).
- **Triage & Confirmation:** The analyst will check the endpoint logs in Wazuh to confirm if `eqig.exe` was executed. The presence of the hidden file in the `AppData` directory confirms a true positive.
- **Scoping:** The analyst will run a SIEM query across all network endpoints looking for the hash of `eqig.exe` or DNS queries to `oekovqlg.org` to ensure no other machines are infected.

3. Containment, Eradication, and Recovery

- **Containment:** The infected Windows 10 endpoint will be immediately isolated from the network using Wazuh's active response capabilities or network firewall rules to prevent C2 communication and lateral movement.
- **Eradication:**
 - Terminate the malicious process (PID identified during analysis).
 - Delete the dropped payload: `C:\Users\Admin\AppData\Roaming\eqig.exe`.
 - Remove the persistence mechanism: Delete the malicious Registry key under `HKCU\Software\Microsoft\Windows\CurrentVersion\Run`.
- **Recovery:** Reconnect the machine to the network and run a full-system anti-malware scan. Monitor the endpoint's traffic for 24 hours to verify that no malicious behavior resumes.

4. Post-Incident Activity

- **Improvements:** Update Suricata rules to include any new DGA domains discovered during the incident. Enforce stricter execution policies (e.g., AppLocker) to prevent unauthorized .exe files from running out of the AppData\Roaming directory.
 - **Documentation:** Compile the incident logs, conduct a lessons-learned meeting, and add the newly discovered hashes to the organization's internal Threat Intelligence platform.
-

Consolidated IOC & IOA Reference Table

Type	Indicator / Behavior	Source	Significance
IOC (Domain)	oekovqlg.org	Network (PCAP)	Zeus DGA Domain for Command & Control (C2) communication.
IOC (File Path)	C:\Users\Admin\AppData\Roaming\eqig.exe	Host (Triage)	Default drop location for the Zeus payload copy.
IOA (Command)	attrib.exe +H +S [filepath]	Host (Triage)	Evasion technique used to hide the malware file from the user.
IOA (Persistence)	Registry write to CurrentVersion\Run	Host (Triage)	Autostart mechanism to ensure malware survives reboots.

Simulated Attack Timeline (Zeus Detonation)

- **T+00:00** - User executes the malicious file ([eqig.ex_](#)).
 - **T+00:02** - Malware copies itself to the [AppData\Roaming](#) directory.
 - **T+00:05** - Malware executes Windows utility [attrib.exe +H +S](#) to hide the newly copied file.
 - **T+00:08** - Malware adds a Run key to the Windows Registry for persistence.
 - **T+00:15** - Malware attempts C2 communication by resolving the DGA domain [oekovqlg.org](#) via port 53 (DNS).
-